COUNTABLE FUNDAMENTAL SYSTEMS OF OPEN IDEALS DESCEND ALONG A TAUT MAP

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement opens with the chapter's phrase 'In the situation above', which fixes the two rings and the taut map between them; that situation is recorded as a standard prerequisite rather than delivered as a context span.

Editorial note. The counted spans carry 2 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Tautness of the given continuous map, and the closure of the extension of an open ideal. Same chapter, the definition of a taut map, in force in the statement and used at proof lines 1809-1814. Produces the chosen open ideals and locates the closure of the extension of an arbitrary open ideal.
- The map between the quotients by an ideal and its extension is faithfully flat in the taut descent situation. Same chapter, the taut descent situation recalled in the statement. Makes the map injective, so that a vanishing after composing with it descends.

Lemma 0.1. *In the situation above, if B has a countable fundamental system of open ideals, then A has a countable fundamental system of open ideals.*

Proof. Choose a fundamental system $B \supset J_1 \supset J_2 \supset \ldots$ of open ideals. By tautness of φ, for every n we can find an open ideal I_n such that $J_n \supset I_n B$. We claim that I_n is a fundamental system of open ideals of A. Namely, suppose that $I \subset A$ is open. As φ is taut, the closure of IB is open and hence contains J_n for some n large enough. Hence $I_n B \subset IB$. Let J be the closure of IB in B. Since $A/I \to B/J$ is faithfully flat, it is injective. Hence, since $I_n \to A/I \to B/J$ is zero as $I_n B \subset IB \subset J$, we conclude that $I_n \to A/I$ is zero. Hence $I_n \subset I$ and we win. $\square$

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